Leadership through own-ad renewal

Can a peer lead repeatedly using only its own advertisements?

Earlier expiration → higher leader priority

Peer X

Publishes a short-lived ad

Waits for inbound followers

→

DHT group key

X: active, expiration E

Later E replaces earlier E

→

Honest peers

Read advertisements

Join an earlier candidate

→

Group

Native admission

Native allreduce

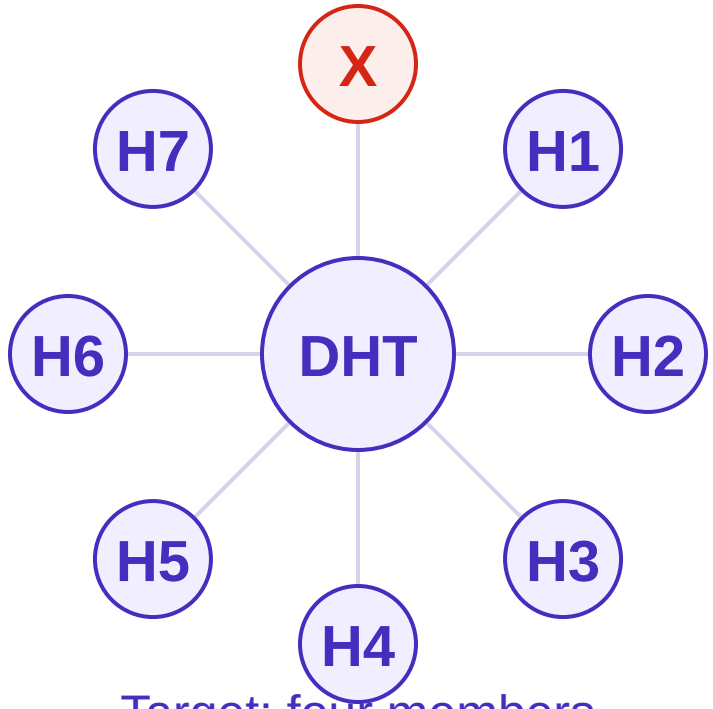
Renew with an increasing expiration: E_new = max(nextafter(E_old, +∞), now + d)

d is the remaining lifetime; f is the requested write frequency.

X changes its own publisher and refuses outbound joins.

Eight peers, controls, repeated rounds

Seven honest peers + X · loopback DHT · tiny tensors



Target: four members

Native admission and allreduce

56 grouping trials

14 policies × 4 seeded arrival scenarios

21 continuous trials

7 policies × 3 new seeds; 30 s windows + drain

Verified completion

Contributor IDs, steps, inputs, every output tensor

X returns after 0.05 s in every continuous arm.

Honest simulated training delay: mean 1.5 s.

Compare stock, passive stock, expiration-only, and periodic refresh.

Short ads lead often; writes add cost

Totals across three 30-second submission windows, including drain

Policy / remaining lifetime	X-led / verified	Ad writes
Stock, minimum four / 5 s	67 / 113	115
Passive stock publisher / 5 s	75 / 101	86
Expiration-only / 1.25 s	106 / 108	143
Refresh 0.5/s / 2.086 s	104 / 108	117
Refresh 4/s / 0.336 s	107 / 110	393

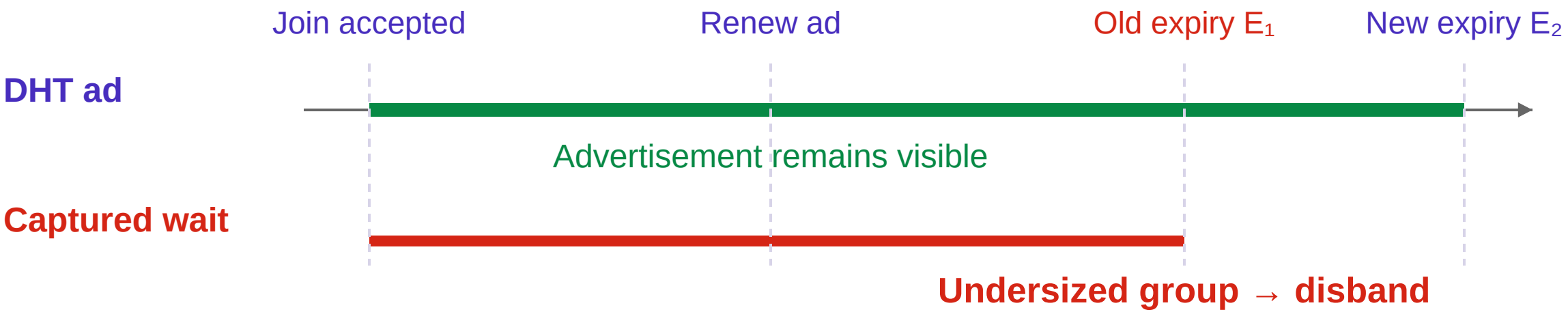
4/s: +3 verified X-led groups; 3.36× ad writes

Five incorrect-tensor steps excluded across the seven tested policies.

Local observations; frequency and lifetime vary together in the refresh comparison.

A visible ad can still lose its group

A later advertisement does not extend an accepted follower's wait.



Disband replies: 9 vs 300

0.5/s vs 4/s across continuous runs

Stream replies, not distinct groups

Continuity: 292/292 active probes

Safe 0.5/s renewal over 30 seconds

40 ms storage-processing delay; no matchmaking

Next: full Optimizer scheduling and routed-delay contention

WAN behavior, longer runs, and the tensor-error cause remain open.